

Nested routes + a dynamic [slug]

Live demo: `/blog` (and `/blog/authors` , `/blog/hello-world`) Code: `src/app/blog/`

What it is

Two concepts in one folder:

1. **Nested routes** — folders nest to nest URL segments, and a parent `layout.tsx` wraps every child.
2. **A dynamic route** — `[slug]` captures a single URL segment as a parameter.

Files in this folder

Path	URL	Notes
<code>blog/layout.tsx</code>	—	Wraps the index, <code>authors</code> , and every <code>[slug]</code>
<code>blog/page.tsx</code>	<code>/blog</code>	Post list
<code>blog/authors/page.tsx</code>	<code>/blog/authors</code>	A nested static route
<code>blog/[slug]/page.tsx</code>	<code>/blog/:slug</code>	Dynamic — reads the slug
<code>blog/_lib/posts.ts</code>	—	Data (private folder, not routable)

The dynamic route — the important details

```
// app/blog/[slug]/page.tsx
export default async function PostPage({
  params,
}): {
  params: Promise<{ slug: string }>; // ← params is a PROMISE in Next 15/16
} {
  const { slug } = await params;    // ← so you await it
  ...
}
```

- **params is a Promise** (Next 15/16). The component is `async` and `await` s it. (In older Next it was a plain object.)
- **generateStaticParams()** pre-renders known slugs at build time (this demo pre-builds `hello-world` , `app-router-basics` , `server-components`). The build log shows them under `/blog/[slug]` .
- **generateMetadata()** builds a per-page `<title>` from the awaited params.
- An unknown slug calls **notFound()** → the nearest `not-found.tsx` (here the root one) renders.

When to use

Dynamic segments for any "detail" page keyed by an id/slug: posts, products, users, etc. Nested folders whenever a URL has sub-sections that should share a layout.

Interview Q&A

- **Q: How do you read a route param in the App Router (Next 15/16)?** `params` is a Promise — make the component `async` and `await params` .

- **Q: What does `generateStaticParams` do?** Tells Next which dynamic values to pre-render at build time (SSG); others render on demand.
- **Q: Does a child route automatically get the parent layout?** Yes — every segment is wrapped by all ancestor layouts.